

Claude Code

A field guide for data scientists

What We Will Cover

- 01 What Claude Code is
- 02 How it differs from chat
- 03 Getting set up
- 04 Everyday data science workflows
- 05 Habits that make it shine



Claude Code

A terminal-based AI agent that reads your project, writes and runs code, inspects the results, and iterates until the job is done.

Chat vs Agent

Chat

You ferry code around

vs

Agent

It works in your repo

What It Can Do in Your Project

Explore

Read files, schemas, and notebooks.

Execute

Run scripts, queries, and tests.

Iterate

Check outputs and fix its own errors.

Set Up in Three Steps

01 Install

`npm install -g @anthropic-ai/claude-code`

02 Open your project

cd into the repo and run claude.

03 Ask for an analysis

Describe the outcome; it plans the code.



a first session

```
$ cd sales-analysis && claude
```

```
> Profile sales.csv: nulls, dtypes,  
  outliers. Save it to profile.md.
```

```
Reading sales.csv (1.2M rows)...
```

```
Running profiling script...
```

```
Wrote profile.md: 3 findings.
```

Everyday Workflows

EDA

Profile new data in minutes.

Cleaning

Fix types, nulls, duplicates.

Modeling

Fast first baselines.

Reporting

Plots and write-ups.

How It Works on a Task

01

Explore

Scans the data and code.

02

Plan

Proposes an approach.

03

Execute

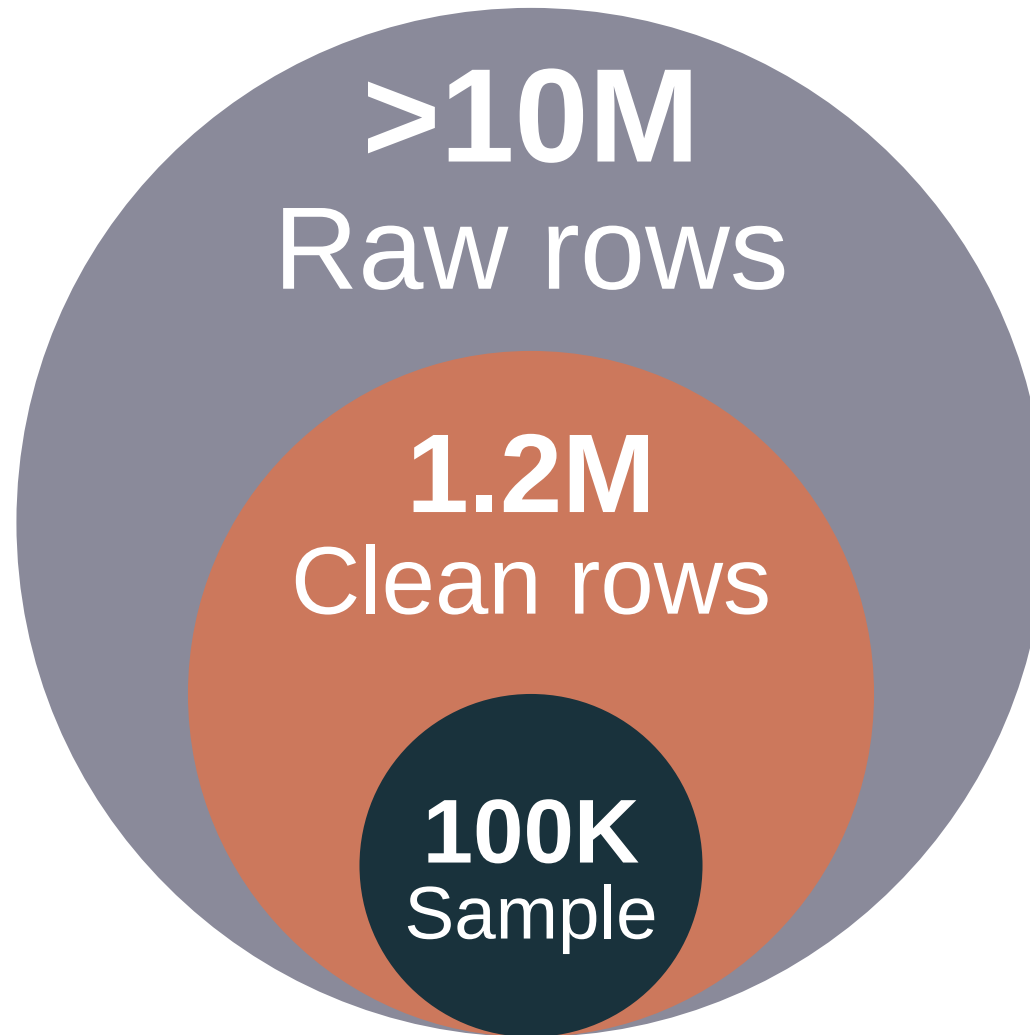
Writes and runs the code.

04

Verify

Checks results, then iterates.

One Prompt, a Whole Pipeline



A Typical Day, Reworked

Before

Boilerplate pandas by hand

Stack Overflow detours

Stale, undocumented
notebooks

With Claude Code

Describe the outcome
instead

Errors debugged in place

Analysis committed with
notes

Commands Worth Knowing

Command

What it does

`claude`

Start a session

`/init`

Write CLAUDE.md

`/compact`

Trim the context

`claude -p`

One-shot mode

Habits That Make It Shine

Keep a CLAUDE.md with data caveats.

Ask for a plan before a big analysis.

Have it verify outputs, not just run.

Commit small, reviewable steps.



**What would you analyze if
the boilerplate wrote itself?**

Start Today

Open a terminal and ask